AgriSense - AI-Powered Smart Agriculture Drone & Soil Health Station

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Abstract

Modern agriculture struggles with one core problem: using resources wisely—especially water—while keeping crops healthy and productive. This project proposes an Autonomous Multi-Tiered Precision Agriculture System that tackles both issues through coordinated aerial and ground-based automation. The goal is simple: detect crop issues early, manage soil, conditions intelligently, and reduce waste without needing constant human oversight. The system operates through two main components that work hand-in-hand.

1. The UAV (Drone) Unit:

This unit performs fast, high-resolution surveys of farmland using onboard processors and specialized imaging sensors. Instead of simply taking pictures, it analyzes them in real time to spot trouble—pest attacks, nutrient shortages, or early signs of disease. By mapping these problem areas across the field, the drone generates a health index that guides where action is needed most.



Figure 1: Flying Drone

2. The Ground Station:

This is the brain of the operation. It collects continuous readings from soil and environmental sensors—tracking parameters like moisture, temperature, and humidity. When the drone

flags a problem zone, the ground station cross-references that data with its own. The system then automatically adjusts irrigation through a segmented network, delivering precise water amounts only to affected areas. This ensures that every section of the field gets exactly what it needs—nothing more, nothing less.

Together, these two layers form an intelligent loop: the drone sees the big picture, and the ground station handles the fine details. The result is smarter water use, faster detection of crop stress, and healthier yields with far less manual effort. This system represents a step towards a more sustainable, data-driven form of farming—where decisions are made by real-time evidence, not guesswork.

Project Objective

The main goal of this project is to design and implement an integrated, autonomous system that merges aerial imaging with on-ground environmental monitoring. The system aims to make farming smarter—using data to manage resources efficiently and detect crop problems before they escalate, ultimately improving yield and sustainability.

Key Aims:

1. **Develop a Vision-Guided UAV Platform**
   Build a drone capable of scanning crop fields automatically using advanced imaging and computer vision. Its purpose is to quickly identify and map problem areas—whether caused by pests, diseases, or nutrient deficiencies—across large sections of farmland.
2. **Create a Ground Station Monitoring and Control Network**
   Set up a ground-based system that continuously tracks key conditions like soil moisture, humidity, and temperature. This station will analyze local sensor data to understand each zone's specific resource needs.
3. **Enable Precision Resource Allocation**
   Combine insights from both aerial and ground data to control irrigation intelligently. Water will be distributed only where and when it's actually needed, cutting waste and maintaining consistent soil and plant health.
4. **Validate System Performance**
   Test and prove the system's real-world impact by measuring its ability to conserve resources (especially water), detect issues early, and improve overall crop productivity and resilience.

This approach transforms raw data into direct, automated action—turning farms into adaptive systems that respond to real-time conditions rather than fixed schedules or guesswork.

Core Features

Soil Monitoring - moisture, temperature, and humidity (real-time)

- The ground station uses soil moisture probes placed in each zone and a DHT22 (or similar) for local air temperature and humidity.
- Readings are sampled periodically and used for both reactive and scheduled irrigation decisions.
- Simple user-friendly readouts are shown on the Edge Panel (if present) and sent to the dashboard when available.



Figure 2: Ground Station setup for soil monitoring

Drone Aerial Crop Scanning - AI-Based Plant Health Detection

- A quadcopter-mounted camera captures overhead images of the crop area.
- The ML model is used from scratch and the details of the model are
    1. Model parameter: 2.5 million
    2. Model validation accuracy: 92.09%
    3. Model type: fine-tuned
    4. Machine Learning Approach: Transfer Learning

5. Model Architecture: Convolutional Neural Network (CNN, MobileNet V2)
6. Model Size: 23.7 MB

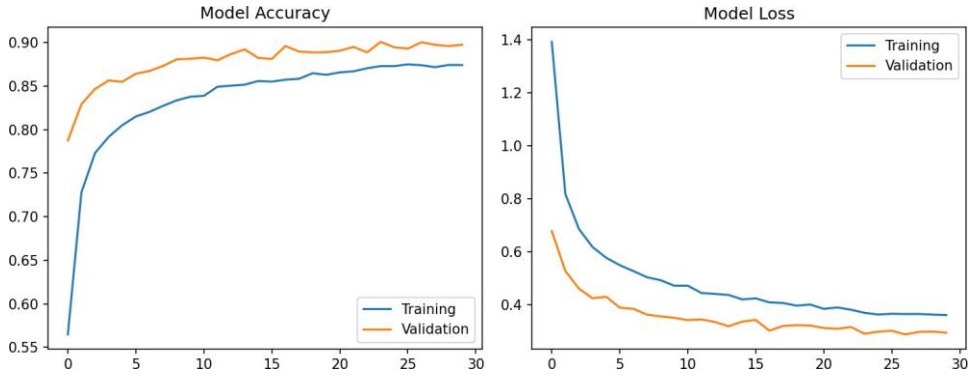


Figure 3: Training Process Visualization

The figure shows the process of the data training.

Automated Irrigation Control - Threshold-Based Watering

- The ground station actuates a water pump through a relay when soil moisture falls below a user-defined threshold.
- Pumped water is delivered to the selected zone via tubing or a drip emitter.
- The system measures moisture after watering and stops automatically when the setpoint is reached.

Multi-Zone Smart Irrigation Manifold

- A servo-actuated manifold (rotary diverter) or multiple solenoid valves route water from a single pump to different zones.
- The Arduino selects the target zone based on sensor readings and drone-flagged areas, allowing for precise, efficient watering.
- This reduces water waste and demonstrates scalable farm deployment using one pump for many zones.

Supporting Component

Wireless Data Visualization - Web Dashboard (Not a main feature but included)

- The dashboard aggregates sensor data, shows the drone feed, displays scheduled and executed irrigation, and allows remote commands.

- It is presented as a supporting subsystem to illustrate the system's usability and to demonstrate data fusion during the proposal presentation.

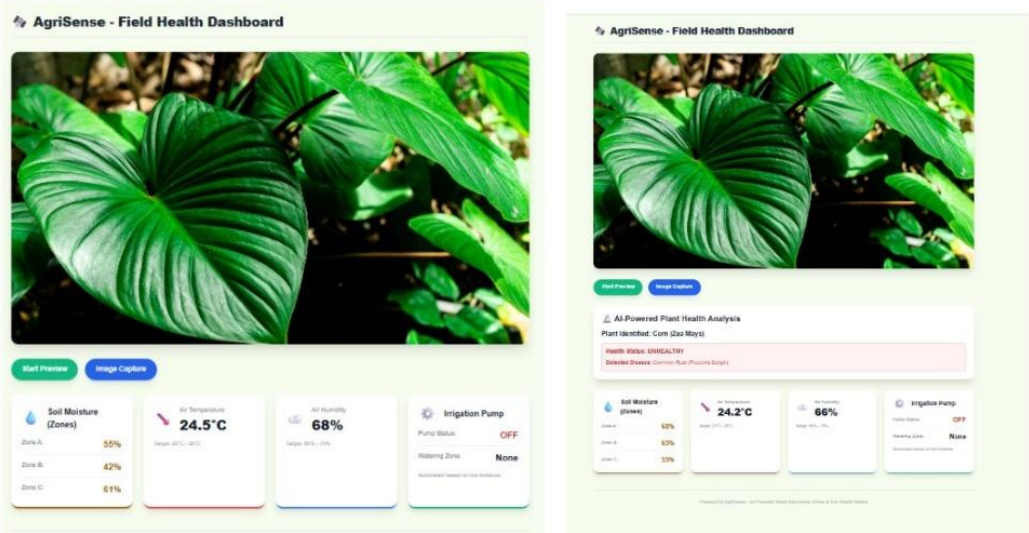


Figure 4: Dashboard

Project Motivation: Building Smarter, Sustainable Agriculture

The pressure on global agriculture keeps rising—more mouths to feed, less water to spare, and increasingly unpredictable weather. Traditional farming practices, which rely on uniform irrigation and delayed response to crop issues, simply can't keep up. Too much water gets wasted, and too many problems are caught too late. This project grows out of the need to change that—to create a smarter, data-driven approach that uses precision instead of prediction.

1. Tackling Resource Waste

A huge portion of water, fertilizer, and pesticides is still applied blindly—spread evenly across fields regardless of what each area actually needs. The goal here is to end that guesswork. By combining drone-based aerial monitoring with real-time soil data from the ground, the system can fine-tune how and where resources are used. The Ground Station's constant tracking of soil moisture and temperature allows irrigation to adjust automatically, zone by zone. It's a practical solution to over-irrigation, especially in regions already struggling with water shortages.

2. Making Intervention Proactive, Not Reactive

In traditional farming, issues like pests or nutrient deficiencies are often noticed only when the damage is visible—and by then, it's usually costly to fix. The UAV component changes that timeline. Using computer vision, it spots early signs of trouble before they spread, pinpointing the exact location of affected areas. This early warning system gives farmers time to act quickly, reducing crop loss, limiting chemical use, and preserving both yield and quality.

3. Pushing Agriculture Toward Data Intelligence

At its core, this project isn't just about automation—it's about insight. By continuously collecting and merging aerial and ground data, the system creates a self-improving feedback loop. Farmers gain real, actionable information about what their fields need and when they need it. That's the foundation of modern, sustainable agriculture: informed decisions driven by live data, not outdated habits.

In short, this project aims to bridge the gap between technology and farming—making agriculture not only more efficient but also resilient enough to thrive in a resource-constrained, climate-affected world.

Components

1. For the Ground Station:

- Arduino UNO
- Jumper
- Bred board
- 4 Channel Relly
- ESP- 32
- DHT-22
- Soil Sensor
- Servo Motor
- 12v 2A charger
- Oled Display I2C

- Pump

2. For the UAV(Drone)

- Raspberry Pi4gb
- Camera V2
- F405 V3 PLUS Flight Controller
- ESC
- MATEK XT 60 PDB
- DJI 2212 920KV VLDX Motor
- Propeller
- F450 Quadcopter Frame
- 3500mAh Lipo Battery
- High Temperature Silicone Wire
- 20CM Strap Battery Binding

Dataset Description

The Plant Village dataset (https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset/data) is a widely used dataset for plant disease classification, containing over 54,000 images of healthy and diseased leaves across 38 categories. It has become one of the benchmark datasets for training and evaluating deep learning models in agricultural disease detection due to its large variety and balanced data distribution.

From this dataset, we selected a subset of 20,639 images to train our model for detecting 15 specific classes. These include:

1. Pepper_bell_Bacterial_spot
2. Pepperbell_healthy
3. Potato_Early_blight
4. Potato_Late_blight
5. Potato_healthy
6. Tomato_Bacterial_spot
7. Tomato_Early_blight
8. Tomato_Leaf_Mold

9. Tomato_Septoria_leaf_spot
10. Tomato_Spider_mites_Two_spotted_spider_mite
11. TomatoTarget_Spot
12. TomatoTomato_YellowLeafCurl_Virus
13. Tomato_Tomato_mosaic_virus
14. Tomato_healthy

This subset was chosen to balance both healthy and diseased categories across major crop types.

Before model training, all dataset and test images were preprocessed using the open-source background remover model **Rembg** (https://github.com/danielgatis/rembg). The backgrounds of the images were removed, and a plain white background was added to ensure visual uniformity and reduce noise from background elements. This preprocessing step helps the model focus more effectively on leaf features relevant to disease detection.

System Architecture Overview

The project is structured as a Multi-Tiered Distributed Network consisting of two major subsystems that communicate wirelessly: the Unmanned Aerial Vehicle (UAV) Tier and the Ground Station Tier. The UAV focuses on macro-scale data acquisition (crop health imaging), while the Ground Station handles micro-scale data acquisition (soil conditions), system decision-making, and physical corrective actions (irrigation).

I. UAV Tier Architecture (Data Acquisition and Transmission)

The drone acts as an autonomous aerial sensor platform. Its primary function is high-resolution data capture and processing to generate a health map of the crops.

II. Ground Station Tier Architecture (Integration, Decision, and Control)

The Ground Station serves as the central hub, fusing aerial data with localized sensor readings to make intelligent decisions about irrigation and environmental control.1 The ESP32 is used for complex logic and Wi-Fi communication, while the Arduino UNO handles stable, non-time-critical sensor polling and actuator control.

Sub-Component	Component Used	Role in System
Input Sensor	**Camera V2**	Captures high-resolution visual data (RGB) of the crops below for subsequent image analysis.
Data Processing Core	**Raspberry Pi 4GB**	Serves as the high-power onboard flight computer. It runs computer vision algorithms (e.g., color-based thresholding or simple machine learning models) on the Camera V2 feed to detect areas of stress or disease and translates these findings into GPS coordinates.
Communication (RPi Wi-Fi)		Uses the Raspberry Pi's built-in Wi-Fi capabilities to send the processed anomaly data (zone ID, coordinates, severity) to the ESP32 module on the ground station.
Flight Control	**F405 V3 PLUS FC, ESC, Motor, Propeller**	This dedicated subsystem manages flight stability, motor speed, and navigational telemetry, operating independently of the RPi's vision processing core.

Detailed Component Descriptions

Input Sensors

Data Processing Logic

The core processing happens in two steps:

1. **UAV Edge Processing (Raspberry Pi 4GB):** The Pi runs a lightweight computer vision script (e.g., using OpenCV) to analyze the Camera V2 frames. It isolates colors that correspond to unhealthy vegetation (browning, yellowing, etc.), calculates the area of anomaly, and packages this data with the corresponding GPS coordinates before transmitting it.
2. **Ground Fusion and Decision (ESP32):** The ESP32 continually receives current sensor values from the Arduino UNO. When it receives an anomaly report from the UAV, it checks the local soil moisture in that anomaly zone. The Decision Logic is simple: IF (Soil Moisture < Threshold) AND (UAV reports Stress in this Zone) THEN Activate Pump for this Zone.

Output Devices and Actuation

Combined Project Components List (Prices in BDT)

This list provides the lowest current estimated retail prices found online in Bangladesh Taka

Sub-Component	Component Used	Role in System
Communication Gateway	**ESP32**	Acts as the primary Wi-Fi module, receiving problem reports from the UAV and publishing local sensor data to a cloud service (if used) or a local dashboard. It also hosts the main decision logic.
Sensor Aggregator	**Arduino UNO**	Provides stable, reliable analog and digital input polling for the low-power sensors. The Arduino passes conditioned, sampled data to the ESP32 for fusion.
Decision Logic	**ESP32 (Primary)**	Fuses the two data streams: UAV Health Map (problem locations) and Local Sensor Readings (environmental conditions). The logic determines if the soil is too dry and if the area is a high-priority stress zone, triggering a watering sequence.
User Interface	**OLED Display I2C**	Provides instant, localized feedback to the operator on current soil moisture, ambient temperature, system status, and the last detected UAV anomaly.
Power Management	**12V 2A Charger, Bred Board, Jumper**	Provides regulated power to all components (sensors, microcontrollers, and pump/relay) and facilitates component interconnection.

(BDT) for the specified components.

Table 1: Combined Project Components List (Prices in BDT)

Component Name	**Sets Required**	**Price per Set (BDT)**	**Subtotal (BDT)**
Automation & IoT Components			
Arduino UNO (SMD CH340)	1	435	435
Jumper Wires (40 Pcs Set)	3	85	255
Breadboard (400 Tie Points)	3	87	261
4 Channel Relay (5V)	1	240	240
ESP-32 (WROOM32)	1	790	790
DHT-22 (Temp/Humidity Sensor)	1	235	235
Soil Sensor	3	150	450
Servo Motor (SG90)	2	180	360
12V 2A Charger/Adapter	1	120	120

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Table 1 – continued from previous page

Component Name	Sets Required	Price per Set (BDT)	Subtotal (BDT)
OLED Display I2C (0.96")	1	379	379
Water Pump (Mini DC 3-6V)	1	63	63
Drone/FPV Components			
Raspberry Pi 4GB	1	9,700	9,700
Camera V2	1	3,390	3,390
F405 V3 PLUS Flight Controller	1	8,499	8,499
ESC (Single Unit, e.g., 30A SimonK)	4	499	1,996
MATEK XT 60 PDB	1	948	948
DJI 2212 920KV VLDX Motor (Single Unit)	1	1,120	1,120
Propeller (1045 Pair)	2	140	280
F450 Quadcopter Frame	1	1,650	1,650
3500mAh Lipo Battery (3S)	1	3,500	3,500
High Temperature Silicone Wire (2ft Set)	2	130	260
20CM Strap Battery Binding (Single Strap)	2	47	94
	TOTAL COUNT & COST		**36 Sets - 35,070 BDT**

Project Execution Timeline Summary

This project came together over five intense weeks—bringing together two demanding areas: hardware integration (Ground Station and Flight Control) and artificial intelligence (Computer Vision and Automation). Here's how it unfolded, step by step.

Week 1: Planning and Procurement

The first week was all about groundwork. You locked down the full project concept, detailing how both the drone and the ground station would interact as one system. Once the plan was solid, it was time to shop—36 different components in total, from Raspberry Pis and flight controllers to sensors, pumps, and frame parts. Getting everything in place early ensured that the build phase could move without delays.

Component	Parameter Measured	Tier Function
Camera V2	Visual/RGB Image Data	UAV: Captures high-definition images for vision-based crop health analysis.
Soil Sensor	Soil Moisture Content	Ground: Measures the volumetric water content in the soil. This is the primary trigger for the irrigation system.
DHT-22	Air Temperature and Humidity	Ground: Monitors the ambient environment. This data is critical for calculating evapotranspiration rates and optimizing the next watering cycle.

Week 2: Ground Station and Hardware Integration

This week was dedicated to getting the automation system alive. You connected the Arduino/ESP-32 with the main sensors and actuators—the 4-Channel Relay, DHT-22, Soil Sensors, and Water Pump. The key milestone here was establishing reliable communication between components. The firmware and logic tests confirmed the system could read environmental data and trigger irrigation automatically—exactly as designed.

Week 3: Core Flight System and First Flight

Focus shifted to the drone. You built out the F450 quadcopter frame, installed the DJI 2212 motors, ESCs, and F405 flight controller, and tuned the firmware. The moment of truth came with the first controlled flight test—which it passed. This verified that the drone's propulsion and control systems were stable and ready for the next phase: intelligence integration.

Week 4: Intelligence and Vision Integration

With the drone's body and flight mechanics working, it was time to give it a brain and eyes. You mounted the Raspberry Pi (4GB) and Camera V2, then configured the software environment. Most of the effort this week went into training and optimizing the machine learning model—teaching the system how to identify target zones, recognize crop stress, and handle real-time data inputs from the camera.

Week 5: Finalization and System Integration

Component	Control Source	Function
4 Channel Relay	ESP32	Acts as an electrical switch, using a low-power digital signal from the ESP32 to safely control the high-power 12V supply required by the pump.
Pump	4 Channel Relay	The final actuator that physically draws and distributes water to the targeted irrigation zone upon command from the relay.
OLED Display I2C	ESP32/Arduino	Provides human-readable feedback on the current status of the sensors (e.g., "Moisture LOW", "Irrigating Zone 3").
Servo Motor	ESP32/Arduino (Potential Use)	Can be used to control a physical valve or a directional nozzle to divert water flow to one of four predetermined irrigation zones, correlating with the 4 channels of the relay.

The final week tied everything together. The flight-tested hardware was merged with the trained AI system, completing the autonomous feedback loop. You refined the build—adjusting wire layouts, battery mounts, and component placement—and then ran full-scale tests. The project wrapped up with a successful demonstration of a fully functional, intelligent drone system capable of autonomous monitoring and resource control.

In five weeks, you took the project from concept to completion—an operational system that combines flight, sensing, and AI into one cohesive precision agriculture platform.

Project Conclusion and Future Scope

Over the course of five weeks, this project successfully brought together two complex fields—embedded automation and intelligent aerial robotics—into one cohesive working system. Each stage was structured with precision: starting from component acquisition and ground station setup, moving through drone assembly and flight testing, and ending with the integration of a computer vision system powered by the Raspberry Pi. The end result—a drone capable of

autonomous mission execution and a separate, fully functional ground monitoring setup—proves both the soundness of the design and the technical depth of the implementation. This achievement lays a solid foundation for future expansion into real-world agricultural use.

Future Improvements and Development Directions

To evolve this prototype into a more capable and reliable system, future work should focus on expanding autonomy, improving communication reliability, and strengthening power management.

1. **Stronger Communication and Ground Control**
   Currently, communication between the drone and the ground station is limited. Upgrading to a two-way telemetry system using high-frequency radio (433 MHz or 900 MHz) would change that. It would enable real-time mission updates, in-flight reprogramming, and continuous transmission of flight and vision data to a control laptop. Turning the Arduino/ESP-32 unit into a dedicated mission control hub would make the system more dynamic and field-manageable.

2. **Smarter Energy Management**
   The drone currently depends on a single LiPo battery, which poses a risk if power fails mid-flight. Introducing a backup power source—such as a secondary battery or super capacitor—for critical components like the Raspberry Pi and Flight Controller would add a layer of safety. On top of that, optimizing the Raspberry Pi's code for lower power use and adding a reliable low-voltage alert would extend flight time and make energy consumption more predictable.

With these refinements, the system could move beyond a prototype into a robust, field-ready agricultural tool—capable of autonomous operation, reliable communication, and safe, efficient energy use in real-world farming conditions.